Ablation performance

Six variants evaluated on the selected graph.

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.700 ± 0.002	0.681 ± 0.003	**0.713 ± 0.001**	0.688 ± 0.005	0.699 ± 0.002	0.689 ± 0.001
10	0.700 ± 0.002	0.682 ± 0.004	**0.706 ± 0.000**	0.688 ± 0.004	0.699 ± 0.001	0.689 ± 0.002
15	0.699 ± 0.002	0.681 ± 0.003	**0.701 ± 0.005**	0.687 ± 0.003	0.698 ± 0.001	0.690 ± 0.002
20	0.698 ± 0.001	0.681 ± 0.002	**0.700 ± 0.003**	0.687 ± 0.003	0.698 ± 0.000	0.689 ± 0.002
25	0.698 ± 0.001	0.682 ± 0.002	**0.698 ± 0.001**	0.687 ± 0.003	0.697 ± 0.000	0.689 ± 0.002
30	0.697 ± 0.001	0.682 ± 0.001	**0.697 ± 0.000**	0.687 ± 0.003	0.697 ± 0.000	0.689 ± 0.002
35	**0.697 ± 0.000**	0.682 ± 0.001	0.696 ± 0.000	0.686 ± 0.003	0.696 ± 0.000	0.689 ± 0.001
40	**0.696 ± 0.000**	0.682 ± 0.001	0.696 ± 0.000	0.686 ± 0.002	0.696 ± 0.000	0.689 ± 0.001
45	**0.696 ± 0.000**	0.683 ± 0.000	0.696 ± 0.000	0.686 ± 0.002	0.696 ± 0.000	0.688 ± 0.001
50	**0.696 ± 0.000**	0.683 ± 0.001	0.695 ± 0.000	0.686 ± 0.002	0.695 ± 0.000	0.688 ± 0.001

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.990 ± 0.002**	0.985 ± 0.003	0.984 ± 0.001	0.986 ± 0.001	0.986 ± 0.002	0.982 ± 0.003
10	**0.989 ± 0.002**	0.986 ± 0.003	0.981 ± 0.000	0.985 ± 0.000	0.987 ± 0.001	0.982 ± 0.003
15	**0.987 ± 0.002**	0.985 ± 0.001	0.984 ± 0.001	0.985 ± 0.001	0.987 ± 0.001	0.981 ± 0.002
20	**0.987 ± 0.001**	0.985 ± 0.001	0.985 ± 0.000	0.984 ± 0.001	0.986 ± 0.001	0.980 ± 0.002
25	0.986 ± 0.001	0.985 ± 0.001	**0.986 ± 0.000**	0.983 ± 0.001	0.984 ± 0.002	0.981 ± 0.001
30	**0.986 ± 0.000**	0.985 ± 0.001	0.986 ± 0.000	0.982 ± 0.001	0.984 ± 0.002	0.981 ± 0.001
35	0.985 ± 0.001	0.984 ± 0.001	**0.986 ± 0.001**	0.982 ± 0.001	0.984 ± 0.002	0.981 ± 0.000
40	0.985 ± 0.000	0.984 ± 0.001	**0.986 ± 0.001**	0.982 ± 0.002	0.984 ± 0.001	0.981 ± 0.001
45	0.984 ± 0.001	0.984 ± 0.001	**0.985 ± 0.001**	0.981 ± 0.002	0.983 ± 0.001	0.982 ± 0.000
50	0.984 ± 0.001	0.984 ± 0.001	**0.985 ± 0.001**	0.981 ± 0.003	0.983 ± 0.001	0.982 ± 0.000

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Comparison performance

Eight baselines and 2CKG4ER evaluated on the selected graph.

(a) Adaptability (Ada)

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	0.688	0.689	0.612	0.689 ± 0.002	0.691 ± 0.002	0.685 ± 0.001	0.687 ± 0.003	0.689 ± 0.002	**0.700 ± 0.002**
10	0.688	0.688	0.616	0.690 ± 0.002	0.692 ± 0.002	0.686 ± 0.000	0.688 ± 0.003	0.690 ± 0.002	**0.700 ± 0.002**
15	0.688	0.688	0.619	0.690 ± 0.001	0.692 ± 0.002	0.686 ± 0.000	0.688 ± 0.002	0.690 ± 0.001	**0.699 ± 0.002**
20	0.688	0.687	0.621	0.690 ± 0.001	0.692 ± 0.002	0.686 ± 0.001	0.688 ± 0.002	0.690 ± 0.001	**0.698 ± 0.001**
25	0.688	0.687	0.624	0.690 ± 0.001	0.692 ± 0.001	0.687 ± 0.001	0.689 ± 0.002	0.690 ± 0.001	**0.698 ± 0.001**
30	0.688	0.687	0.626	0.689 ± 0.001	0.692 ± 0.001	0.687 ± 0.001	0.689 ± 0.002	0.690 ± 0.001	**0.697 ± 0.001**
35	0.688	0.687	0.628	0.689 ± 0.001	0.691 ± 0.001	0.687 ± 0.001	0.689 ± 0.002	0.690 ± 0.001	**0.697 ± 0.000**
40	0.688	0.687	0.630	0.689 ± 0.001	0.691 ± 0.001	0.687 ± 0.001	0.689 ± 0.001	0.690 ± 0.001	**0.696 ± 0.000**
45	0.688	0.687	0.632	0.689 ± 0.001	0.691 ± 0.001	0.688 ± 0.001	0.689 ± 0.001	0.690 ± 0.000	**0.696 ± 0.000**
50	0.688	0.687	0.633	0.689 ± 0.001	0.691 ± 0.001	0.688 ± 0.001	0.689 ± 0.001	0.690 ± 0.000	**0.696 ± 0.000**

(b) Novelty (NOV)

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	0.986	0.980	0.999	0.999 ± 0.000	0.999 ± 0.001	**0.999 ± 0.000**	0.997 ± 0.001	0.999 ± 0.001	0.990 ± 0.002
10	0.988	0.983	**0.999**	0.999 ± 0.000	0.998 ± 0.001	0.999 ± 0.000	0.997 ± 0.001	0.999 ± 0.001	0.989 ± 0.002
15	0.986	0.982	**0.999**	0.999 ± 0.001	0.998 ± 0.001	0.999 ± 0.000	0.997 ± 0.001	0.998 ± 0.001	0.987 ± 0.002
20	0.985	0.981	**0.999**	0.998 ± 0.001	0.997 ± 0.002	0.999 ± 0.001	0.997 ± 0.001	0.998 ± 0.001	0.987 ± 0.001
25	0.985	0.982	**0.999**	0.997 ± 0.001	0.997 ± 0.002	0.999 ± 0.000	0.997 ± 0.001	0.998 ± 0.001	0.986 ± 0.001
30	0.985	0.981	**0.999**	0.996 ± 0.001	0.997 ± 0.002	0.999 ± 0.000	0.997 ± 0.001	0.998 ± 0.001	0.986 ± 0.000
35	0.984	0.981	**0.999**	0.996 ± 0.001	0.996 ± 0.001	0.999 ± 0.000	0.997 ± 0.000	0.998 ± 0.001	0.985 ± 0.001
40	0.984	0.981	**0.999**	0.995 ± 0.001	0.996 ± 0.001	0.999 ± 0.000	0.997 ± 0.000	0.998 ± 0.001	0.985 ± 0.000
45	0.984	0.981	**0.999**	0.995 ± 0.001	0.996 ± 0.001	0.999 ± 0.000	0.997 ± 0.000	0.998 ± 0.001	0.984 ± 0.001
50	0.984	0.981	**0.999**	0.995 ± 0.001	0.995 ± 0.001	0.999 ± 0.000	0.997 ± 0.000	0.998 ± 0.001	0.984 ± 0.001

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.